



Day 11 – Introduction to SwiftUI

Today I started learning **SwiftUI** and built my first interactive UI using:

- `NavigationView`
- `Form`
- `Section`
- `TextField`
- `Picker`
- `Buttons`
- `State Management`
- `FocusState`
- `Group`
- `VStack`

💎 What I Learned

- Built UI using SwiftUI views
- Used `@State` for dynamic data
- Created input fields using `TextField`
- Used `Picker` for selection menus
- Learned Form & Section layout
- Used `Button` actions
- Learned `NavigationView`
- Used `@FocusState` for keyboard handling
- Displayed dynamic text updates
- Used `Group` and `VStack` for organizing UI

💎 NavigationView

`NavigationView` provides navigation structure and titles.

```
```swift
NavigationView {

}
.navigationTitle("SwiftUI Learning❤️")
```

---

## 💎 Form & Section

Forms help organize user input.

```
Form {

 Section("Your Details") {

 }

 Section("Money 💰") {
```

```
}
}
```

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## ◆ @State Property Wrapper

@State allows SwiftUI to update UI automatically when data changes.

```
@State private var name = ""
@State private var tapCount = 0
```

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## ◆ TextField

Used for user text input.

```
TextField("Enter your Name", text: $name)
```

Dynamic Output:

```
Text("Your name is \(name)")
```

---

## ◆ Picker

Used for selecting values from multiple options.

```
Picker("Select a Warrior", selection: $selectedwarrior) {
 ForEach(warriors, id: \.self) {
 Text($0)
 }
}
```

---

## ◆ Button

Buttons perform actions when tapped.

```
Button("Tap Count: \(tapCount)") {
 tapCount += 1
}
```

---

## ◆ FocusState

Used to manage keyboard focus.

```
@FocusState private var amountIsFocused: Bool
```

Usage:

```
.focused($amountIsFocused)
```

---

## ◆ Currency TextField

```
TextField(
 "Amount",
 value: $checkAmount,
 format: .currency(code: Locale.current.currency?.identifier ?? "INR")
)
.keyboardType(.decimalPad)
```

---

## ◆ Group & VStack

Used for organizing multiple views together.

```
Group {
 VStack {
 Image(systemName: "heart")
 Text("Yo, Bro Time to start SwiftUI")
 }
}
```

---

## ◆ SF Symbols

SwiftUI uses SF Symbols for system icons.

```
Image(systemName: "heart")
```

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## Important Interview Questions

### **1** What is SwiftUI?

SwiftUI is Apple's declarative UI framework used to build iOS applications.

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### **2** What does `@State` do in SwiftUI?

`@State` stores local mutable data and automatically refreshes the UI when values change.

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### 3 What is a `Form` in SwiftUI?

A `Form` is a container used for grouped user input controls.

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### 4 What is the difference between `VStack` and `Group`?

<code>VStack</code>	<code>Group</code>
Adds vertical layout	Only groups views
Affects UI layout	No layout changes
Adds spacing	Invisible container

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### 5 What is `Picker` in SwiftUI?

`Picker` allows users to select values from a list of options.

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### 6 Why do we use `$` in SwiftUI?

`$` creates a `Binding`.

Example:

```
$text
```

SwiftUI uses bindings to update values directly.

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### 7 What is `@FocusState`?

`@FocusState` controls keyboard focus and input handling.

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## Quick Revision

- SwiftUI = Declarative UI framework

- `@State` = Reactive UI updates
  - `TextField` = User input
  - `Picker` = Selection menu
  - `Button` = User interaction
  - `Form` = Structured input layout
  - `Group` = Logical grouping
  - `VStack` = Vertical layout
  - `$` = Binding
  - `NavigationView` = Navigation container
- 



## Day 11 Summary

Today I built my first interactive SwiftUI screen using:

- Forms
- Sections
- Pickers
- Buttons
- State management
- Dynamic text updates
- Keyboard focus handling
- SwiftUI layouts